

method for purification of the recombinant human granulocyte stimulating factor in the milk

Thought for a few seconds

Here is a **practical purification workflow** for recombinant human granulocyte colony-stimulating factor (rhG-CSF) produced in milk.

Overview

Milk is a difficult starting material because it contains:

- casein micelles
- fat/lipids
- whey proteins such as β -lactoglobulin and α -lactalbumin
- proteases
- salts and other small molecules

So purification usually follows this order:

1. remove fat
2. remove casein
3. clarify whey fraction
4. capture rhG-CSF
5. polish to high purity
6. sterile filtration and formulation

Typical purification method

1. Collect and keep milk cold

- Collect fresh milk.
- Add protease inhibitors if allowed in the process.
- Keep at 4°C to reduce protein degradation.

This is important because rhG-CSF can be damaged by proteases in milk.

2. Remove fat by centrifugation

Use refrigerated centrifugation.

- Spin milk at about 4,000–10,000 × g
- Temperature: 4°C
- Remove the top cream/fat layer

Result:

- skimmed milk fraction remains

3. Remove casein

Casein is one of the biggest impurities.

A common way is **acid precipitation at the isoelectric point of casein**.

Method

- Adjust pH slowly to about 4.6 using HCl or acetic acid
- Stir gently at 4°C
- Centrifuge again

Result:

- **casein pellet** forms
- **whey supernatant** contains soluble proteins, including rhG-CSF if it remains soluble at this pH

Alternative

Instead of acid precipitation, some processes use:

- **microfiltration**
- **calcium-induced precipitation**
- **rennet treatment**

But acid precipitation is the most common basic method.

4. Clarify the whey fraction

After removing casein:

- Filter through coarse filter
- Then pass through **0.45 µm** membrane
- Sometimes also **0.22 µm** later

This removes remaining particles before chromatography.

5. Concentrate and exchange buffer

Use **ultrafiltration/diafiltration (UF/DF)**.

- Choose membrane cutoff appropriate for rhG-CSF
- Concentrate the whey fraction
- Exchange into chromatography starting buffer

This step helps reduce volume and prepare the sample for capture.

6. Capture rhG-CSF by chromatography

rhG-CSF is a relatively basic protein, so **ion-exchange chromatography** is often the first main purification step.

Option A: Cation-exchange chromatography

This is often a strong choice if rhG-CSF is positively charged under the loading pH.

Example

- Buffer pH below the protein's pI
- Load onto **SP-Sepharose** or similar cation exchanger
- Wash away weakly bound milk proteins
- Elute with salt gradient

Option B: Heparin affinity chromatography

G-CSF can show affinity to heparin-like ligands in some setups.

- Load clarified sample
- Wash impurities
- Elute with increasing salt

Option C: Immunoaffinity chromatography

If very high selectivity is needed:

- anti-G-CSF antibody column

This gives strong purification, but it is expensive and less common for scale-up.

7. Intermediate purification / polishing

After capture, one or more polishing steps are usually added.

Common choices

a. Reverse-phase HPLC

Useful for very high purity at analytical or small scale.

- Strong purification power
- But can risk denaturation if conditions are harsh

b. Size-exclusion chromatography

Separates:

- aggregates
- fragments
- remaining contaminants

Usually better as a polishing step, not for large-volume capture.

c. Anion-exchange chromatography

Useful in flow-through or bind-elute mode to remove host milk proteins, DNA, endotoxin-like impurities, or charge variants depending on process design.

d. Hydrophobic interaction chromatography

Can help remove remaining contaminants if needed.

8. Viral and bioburden reduction

For biopharmaceutical-grade material, include appropriate safety steps such as:

- low bioburden processing
 - nanofiltration if applicable
 - validated viral clearance strategy
 - sterile 0.22 μm filtration
-

9. Final formulation

Exchange purified rhG-CSF into formulation buffer, for example:

- acetate or phosphate buffer
- salt for stability
- stabilizer such as sugar or polysorbate if appropriate

Then store refrigerated or frozen depending on the formulation.

Example lab-scale workflow

A simple example process could be:

1. Fresh milk collection at 4°C
2. Centrifuge to remove fat
3. Adjust to pH 4.6 to precipitate casein
4. Centrifuge and collect whey
5. Filter whey
6. Concentrate by ultrafiltration
7. Load on cation-exchange column
8. Elute rhG-CSF with NaCl gradient
9. Polish with size-exclusion or RP-HPLC
10. Sterile filter and formulate

Important process concerns

1. Stability of rhG-CSF

You need to confirm:

- pH stability
- temperature stability
- sensitivity to proteases
- aggregation behavior

2. Glycosylation

If rhG-CSF is produced in mammary gland secretion, it may have a **different glycosylation pattern** from bacterial G-CSF.

That can affect:

- bioactivity
- stability
- immunogenicity
- purification behavior

3. Recovery after casein removal

You must verify that rhG-CSF does not precipitate or get trapped during casein removal.

4. Analytical confirmation

At each stage, usually test with:

- SDS-PAGE
- Western blot
- ELISA
- HPLC
- mass spectrometry
- bioactivity assay

Very concise process flow

```
Milk
→ defatting
→ casein removal
→ whey clarification
→ ultrafiltration/diafiltration
→ ion-exchange capture
→ polishing chromatography
→ sterile filtration
→ formulation
```

Best practical starting method

For most cases, the best starting purification strategy is:

- centrifugation for fat removal
- acid precipitation at pH 4.6 for casein removal
- ultrafiltration/diafiltration
- cation-exchange chromatography
- size-exclusion or RP-HPLC polishing

That is the most standard and realistic foundation.

If you want, I can give you a **full step-by-step experimental protocol** with **buffer examples**, **pH values**, and **chromatography conditions**.

sure

Thought for a couple of seconds

Below is a **research-scale example protocol** for purifying **recombinant human G-CSF (rhG-CSF)** from **milk**.

Goal

Isolate rhG-CSF from milk by removing:

- fat
- casein
- major whey contaminants
- aggregates and fine impurities

and obtain a purified protein fraction for **analytical or research use**.

Example purification protocol

A. Materials

Reagents

- Fresh milk sample containing rhG-CSF
- HCl, dilute
- NaOH, dilute
- Sodium acetate
- Sodium phosphate
- NaCl
- Protease inhibitor cocktail, if compatible
- Sterile water
- Optional: glycerol, trehalose, polysorbate 20 or 80 for formulation

Buffers

1. Casein precipitation adjustment

- dilute HCl for pH adjustment to 4.6

2. Cation-exchange equilibration buffer

- 20 mM sodium acetate, pH 4.5 to 5.0

3. Cation-exchange wash buffer

- same as equilibration buffer, optionally with 50 to 100 mM NaCl

4. Cation-exchange elution buffer

- 20 mM sodium acetate, pH 4.5 to 5.0 + 1 M NaCl
- use as gradient or step elution buffer

5. SEC / polishing buffer

- 20 mM sodium phosphate, 150 mM NaCl, pH 7.0 to 7.4

6. Final formulation buffer

Example:

- 10 mM sodium acetate, 150 mM NaCl, pH 4.0 to 5.5
- or
- 10 to 20 mM phosphate, 150 mM NaCl, pH 6.5 to 7.4

Actual final buffer depends on your stability data.

B. Equipment

- Refrigerated centrifuge
 - pH meter
 - magnetic stirrer
 - filtration setup
 - ultrafiltration unit
 - chromatography system
 - cation-exchange column, such as **SP Sepharose**
 - optional size-exclusion column
 - SDS-PAGE and Western blot tools
 - ELISA or activity assay for rhG-CSF
-

Step-by-step procedure

1. Milk collection and pre-handling

- Collect milk into clean cold containers.
- Keep sample at 4°C.
- Add compatible protease inhibitors if your workflow allows.
- Process as soon as possible.

Notes

Milk proteases and microbial growth can reduce recovery. Keep everything cold.

2. Defatting

Procedure

- Transfer milk into centrifuge bottles.
- Centrifuge at 5,000 to 10,000 × g, 20 to 30 min, 4°C.
- Remove the top fat layer carefully.
- Collect the lower skimmed milk phase.

Output

- defatted skim milk
-

3. Remove casein by isoelectric precipitation

Casein precipitates near pH 4.6.

Procedure

- Stir the skimmed milk gently in an ice bath or cold room.
- Adjust pH slowly to 4.6 using dilute HCl.
- Stir for 15 to 30 min at 4°C after reaching pH 4.6.
- Centrifuge at 8,000 to 12,000 × g for 20 to 30 min at 4°C.
- Collect the supernatant carefully.

Output

- acid whey fraction

Important

At this step, check whether rhG-CSF stays soluble.

Take samples from:

- starting milk
- casein pellet
- whey supernatant

and test by **ELISA** or **Western blot**.

If too much rhG-CSF goes into the pellet, you may need to optimize:

- pH
 - hold time
 - salt concentration
 - alternative casein removal strategy
-

4. Clarification

Procedure

Filter the whey supernatant in stages:

- coarse filter or gauze first
- then **0.45 µm**
- optionally **0.22 µm** before column loading if viscosity and fouling are manageable

Output

- clarified whey
-

5. Concentration and buffer exchange by ultrafiltration

This reduces volume and prepares sample for ion exchange.

Procedure

- Use UF with a membrane in a suitable MWCO range, commonly around **10 kDa**
- Concentrate the clarified whey
- Diafilter into **20 mM sodium acetate, pH 4.5 to 5.0**

Suggested endpoint

- reduce volume to a manageable load volume
- conductivity should be low enough for cation-exchange binding

Output

- concentrated, low-salt sample in binding buffer
-

6. Capture step: cation-exchange chromatography

rhG-CSF is commonly purified well by cation exchange under acidic conditions.

Column

- SP Sepharose or equivalent strong cation exchanger

Example setup

- Equilibrate with **20 mM sodium acetate, pH 4.5 to 5.0**
- Load sample
- Wash with equilibration buffer
- optionally wash with **20 mM sodium acetate + 50 to 100 mM NaCl**
- Elute with a **0 to 1 M NaCl gradient** or step gradient

Example procedure

1. Equilibrate column with **5 to 10 CV**
2. Load sample at moderate flow rate
3. Wash with **5 to 10 CV**
4. Elute with linear gradient over **10 to 20 CV**
5. Collect fractions

Fraction analysis

Check fractions by:

- UV absorbance
- SDS-PAGE
- Western blot
- ELISA

Pool fractions containing rhG-CSF.

7. Intermediate concentration

After pooling the cation-exchange fractions:

- concentrate pooled rhG-CSF fractions by UF
- exchange into polishing buffer if needed

Example buffer:

- 20 mM sodium phosphate, 150 mM NaCl, pH 7.0
-

8. Polishing step

Option A: Size-exclusion chromatography

Good for removing:

- aggregates
- fragments
- remaining high or low molecular weight impurities

Procedure

- Equilibrate SEC column in polishing buffer
- Load concentrated pool
- collect monomer peak fractions
- analyze by SDS-PAGE and SEC profile

Option B: Reverse-phase HPLC

Useful at analytical or small scale when very high purity is needed, but it can be harsher.

Option C: Anion exchange in flow-through mode

Useful if contaminants bind and rhG-CSF does not, depending on pH/conductivity.

For most lab-scale purification, **cation exchange + SEC** is a very reasonable combination.

9. Final sterile filtration

- Filter the final pooled product through **0.22 µm** if compatible with your protein recovery requirements
-

10. Formulation and storage

Example research storage buffer

- 10 mM sodium acetate
- 150 mM NaCl
- pH 4.5 to 5.5

Optional stabilizers after confirming compatibility:

- 5 to 10% glycerol
- low concentration trehalose or sucrose
- trace surfactant such as polysorbate

Storage

- short term: 4°C
- long term: aliquot and store at -80°C
- avoid repeated freeze-thaw cycles

Example full workflow summary

Milk

- cold handling
- centrifuge to remove fat
- adjust to pH 4.6
- centrifuge to remove casein
- collect whey
- 0.45 µm filtration
- ultrafiltration/diafiltration into acetate buffer
- SP cation-exchange chromatography
- collect rhG-CSF fractions
- concentrate
- size-exclusion chromatography
- sterile filtration
- formulation/storage

In-process analytical testing

You should test at key stages.

Recommended checkpoints

After defatting

- total protein
- rhG-CSF ELISA

After casein precipitation

- pellet vs supernatant recovery
- SDS-PAGE
- Western blot

After cation exchange

- purity profile
- fraction mapping
- ELISA or activity assay

After polishing

- final purity
 - aggregate level
 - identity confirmation
 - bioactivity
-

Typical assays

- SDS-PAGE
 - Western blot using anti-human G-CSF
 - ELISA for human G-CSF
 - SEC-HPLC for aggregate analysis
 - RP-HPLC if available
 - mass spectrometry
 - cell-based bioactivity assay
-

Important optimization points

1. Recovery during acid precipitation

This is the first thing to verify.

Some rhG-CSF may:

- precipitate
- stick to casein
- lose activity at low pH if exposure is too long

So optimize:

- pH between 4.4 and 5.0
- hold time
- ionic strength
- temperature

2. Binding to cation exchanger

Optimize:

- loading pH
- conductivity
- flow rate
- elution salt gradient

3. Aggregation

Watch for aggregation during:

- acid treatment
- concentration
- freeze-thaw
- high protein concentration

4. Glycosylation

Milk-derived rhG-CSF may be glycosylated differently from bacterial forms, which can affect:

- pI
 - chromatography behavior
 - apparent molecular weight
 - stability
-

Example starting conditions to try

These are reasonable first-pass screening conditions:

Screen 1

- casein precipitation at pH 4.6
- hold 20 min at 4°C
- bind to SP column at 20 mM acetate pH 4.8

- elute with 0 to 1 M NaCl gradient

Screen 2

- casein precipitation at pH 4.8
- shorter hold
- same SP workflow

Screen 3

- compare acid precipitation vs microfiltration for casein removal

Then compare:

- recovery
- purity
- activity

Expected logic of impurity removal

Step	Main purpose
Defatting	remove cream/lipids
Casein precipitation	remove bulk casein
Clarification	remove particles
UF/DF	reduce volume and adjust buffer
Cation exchange	capture rhG-CSF from whey proteins
SEC	remove aggregates/fragments and polish

Practical caution

This protocol is a **research purification template**, not a validated GMP manufacturing process.
For therapeutic use, you would need full validation for:

- viral safety
 - impurity clearance
 - reproducibility
 - sterility
 - potency
 - stability
 - regulatory compliance
-

Simple lab version

If you want the shortest workable version:

1. Centrifuge milk at 4°C to remove fat
 2. Adjust skim milk to pH 4.6
 3. Centrifuge to remove casein
 4. Filter whey
 5. Concentrate and exchange into acetate buffer
 6. Purify on SP-Sepharose
 7. Polish by SEC
 8. Analyze by SDS-PAGE, Western blot, and ELISA
-

I can also give you a **full SOP-style version with exact volumes, example column sizes, and buffer recipes for 1 liter of milk.**

sure

Thought for a couple of seconds

Below is an **SOP-style research protocol** for purifying **recombinant human G-CSF (rhG-CSF)** from **1 liter of milk**.

Scope

This is a **lab/research-use protocol**, not a GMP manufacturing batch record.

It is meant as a **starting method** that you can optimize based on:

- expression level in milk
- glycosylation state
- stability of your rhG-CSF
- available chromatography system and column size

SOP-Style Protocol

Purification of rhG-CSF from 1 L milk

1. Objective

To isolate and enrich recombinant human granulocyte colony-stimulating factor from milk by:

- defatting
- casein removal
- clarification
- concentration/buffer exchange
- cation-exchange capture
- size-exclusion polishing

2. Principle

Milk contains major impurities including:

- fat globules
- casein micelles
- abundant whey proteins

This protocol separates rhG-CSF by:

- centrifugal fat removal
- casein precipitation near pH 4.6
- ultrafiltration/diafiltration
- cation-exchange chromatography
- size-exclusion chromatography

Since rhG-CSF is commonly a **basic protein**, it often binds well to a **strong cation exchanger** at mildly acidic pH.

3. Starting material

- **1 L fresh milk** containing rhG-CSF

- Keep sample at 4°C
 - Process as soon as possible after collection
-

4. Reagents

A. General reagents

- Hydrochloric acid, 1 N
- Sodium hydroxide, 1 N
- Protease inhibitor cocktail, optional and compatible with downstream use
- Sterile distilled water

B. Buffer chemicals

- Sodium acetate trihydrate
 - Sodium phosphate
 - Sodium chloride
-

5. Buffer recipes

Buffer 1: Cation-exchange equilibration / loading buffer

20 mM sodium acetate, pH 4.8

For 1 L:

- Sodium acetate trihydrate: about 2.72 g
- Adjust pH to 4.8
- Bring to 1 L with water
- Filter if needed

Buffer 2: Cation-exchange wash buffer

20 mM sodium acetate, 100 mM NaCl, pH 4.8

For 1 L:

- Sodium acetate trihydrate: about 2.72 g
- NaCl: 5.84 g
- Adjust pH to 4.8
- Bring to 1 L

Buffer 3: Cation-exchange elution buffer

20 mM sodium acetate, 1 M NaCl, pH 4.8

For 1 L:

- Sodium acetate trihydrate: about 2.72 g
- NaCl: 58.44 g
- Adjust pH to 4.8
- Bring to 1 L

Buffer 4: SEC / polishing buffer

20 mM sodium phosphate, 150 mM NaCl, pH 7.2

For 1 L:

- Prepare 20 mM phosphate buffer
- Add NaCl: 8.77 g
- Adjust pH to 7.2

Buffer 5: Example storage buffer

10 mM sodium acetate, 150 mM NaCl, pH 4.5

For 1 L:

- Sodium acetate trihydrate: about **1.36 g**
 - NaCl: **8.77 g**
 - Adjust pH to 4.5
-

6. Equipment

- Cold room or ice bath
 - Refrigerated centrifuge
 - pH meter
 - Magnetic stirrer
 - Filtration setup
 - 0.45 µm filter
 - Optional 0.22 µm filter
 - Ultrafiltration device, around **10 kDa MWCO**
 - FPLC or equivalent chromatography system
 - Strong cation-exchange column, such as **SP Sepharose**
 - Size-exclusion column, optional but recommended
 - SDS-PAGE setup
 - Western blot or ELISA setup
-

7. Procedure

Step 1. Pre-cooling and sample preparation

1. Place milk on ice or at 4°C.
2. Optionally add protease inhibitor.
3. Mix gently.

Record

- sample volume
 - sample appearance
 - collection time
 - storage time before processing
-

Step 2. Defatting

1. Transfer the **1 L milk** into centrifuge bottles.
2. Centrifuge at **5,000 × g** for **30 min** at **4°C**.
3. Remove the top cream/fat layer carefully with spatula or pipette.
4. Recover the lower skim milk fraction.

Expected result

From 1 L milk, you may recover around **850–950 mL** skimmed fraction depending on fat content.

Sample to keep

Take:

- **1 mL whole milk**
- **1 mL skim milk**

for later analysis.

Step 3. Casein precipitation

1. Put skim milk in a cold beaker with gentle stirring.
2. Adjust pH slowly to **4.6** using **1 N HCl**.
3. Add acid dropwise while monitoring pH continuously.
4. After reaching **pH 4.6**, stir gently for **20 min at 4°C**.
5. Transfer to centrifuge bottles.
6. Centrifuge at **10,000 × g for 30 min at 4°C**.
7. Collect the supernatant carefully. Avoid disturbing the pellet.

Output

- pellet: casein-rich fraction
- supernatant: acid whey fraction

Expected volume

Usually around **700–850 mL** supernatant, but it varies.

Important

Save:

- **1 mL whey supernatant**
- small piece or suspension of casein pellet

to measure whether rhG-CSF is being lost into the pellet.

Step 4. Clarification

1. Pass the whey through a coarse filter first if needed.
2. Then filter through **0.45 µm** membrane.
3. If the sample is clear enough and filterable, optionally follow with **0.22 µm**.

Output

Clarified whey suitable for UF and column loading.

Step 5. Concentration and diafiltration

Use a **10 kDa MWCO ultrafiltration membrane**.

1. Load clarified whey into the UF system.
2. Concentrate to around **50–100 mL**.
3. Add **Buffer 1** and continue diafiltration.
4. Perform about **5–8 diavolumes** into **20 mM sodium acetate pH 4.8**.
5. Final sample conductivity should be low enough for cation-exchange binding.

Target

Final volume: about **50 mL**

Record

- starting volume
 - final volume
 - conductivity
 - pH
-

Step 6. Prepare cation-exchange column

Example column

- **SP Sepharose**

- Example lab column volume: **5–20 mL bed volume**

For a 1 L milk prep, a **5 mL** column may work for pilot screening, but a **10–20 mL** column is more comfortable if expression level is high.

1. Equilibrate the column with **5–10 column volumes (CV)** of **Buffer 1**.
-

Step 7. Load sample onto cation-exchange column

1. Confirm sample pH is about **4.8**.
2. Load sample at a moderate flow rate.
3. Collect flow-through for analysis.

Watch for

If rhG-CSF does not bind well, check:

- pH too high
 - conductivity too high
 - protein overloaded
-

Step 8. Wash column

1. Wash with **5–10 CV** of **Buffer 1**
2. Then wash with **5 CV** of **Buffer 2**
This helps remove weakly bound contaminants.

Collect wash fractions if you want to check losses.

Step 9. Elute rhG-CSF

Use either a gradient or step elution.

Option A: Linear gradient

- 0 to 100% **Buffer 3** over **15–20 CV**

Option B: Step elution

Try:

- 20% **Buffer 3**
- 40% **Buffer 3**
- 60% **Buffer 3**
- 100% **Buffer 3**

Collect fractions, for example:

- **2–5 mL per fraction**

Expected

rhG-CSF should elute in salt-containing fractions depending on its exact charge behavior.

Step 10. Analyze capture fractions

Analyze:

- flow-through
- wash
- elution fractions

using:

- **SDS-PAGE**
- **Western blot**

- ELISA

Pool fractions containing rhG-CSF with the best purity.

Typical pooled volume

Often around **10–40 mL**, depending on your column and fractionation.

Step 11. Concentrate pooled fractions

1. Use UF again to concentrate pooled fractions to around **1–5 mL**
 2. Exchange into **Buffer 4** for SEC
-

Step 12. Size-exclusion chromatography polishing

Example

Use a suitable SEC column, such as:

- Superdex-type column appropriate for proteins in this MW range
1. Equilibrate SEC column with **Buffer 4**
 2. Inject concentrated pooled sample
 3. Elute and collect fractions across the monomer peak
 4. Analyze fractions by SDS-PAGE and/or SEC trace

Purpose

Remove:

- aggregates
 - fragments
 - remaining contaminants
-

Step 13. Final pooling

Pool the fractions corresponding to the main monomeric rhG-CSF peak.

Measure:

- protein concentration
 - purity
 - identity
 - activity if possible
-

Step 14. Final sterile filtration

If protein recovery allows:

1. Pass pooled sample through **0.22 µm** sterile filter

This is recommended for clean final handling, though it can reduce yield slightly at very low protein concentrations.

Step 15. Formulation and storage

Exchange into storage buffer if needed.

Example storage buffer

- 10 mM sodium acetate
- 150 mM NaCl
- pH 4.5

Storage

- short-term: 4°C
- long-term: aliquots at -80°C

Avoid repeated freeze-thaw cycles.

8. Example yield tracking table

You should track mass balance like this:

Step	Volume	Total protein	rhG-CSF amount	Recovery %	Notes
Whole milk	1000 mL			100	start
Skim milk					after defatting
Whey supernatant					after casein removal
UF retentate					before column
CEX pool					after capture
SEC pool					final

9. In-process checkpoints

Checkpoint A: After casein precipitation

Test both:

- whey supernatant
- casein pellet extract

This tells you whether pH 4.6 is causing loss.

Checkpoint B: After cation exchange

Run SDS-PAGE and Western blot on:

- load
- flow-through
- wash
- eluate

This tells you whether the protein bound and where it eluted.

Checkpoint C: Final product

Check:

- purity
- concentration
- bioactivity
- aggregation

10. Recommended assays

Identity

- Western blot with anti-human G-CSF
- mass spectrometry if available

Quantitation

- human G-CSF ELISA
- UV absorbance if purified enough

Purity

- reducing and non-reducing SDS-PAGE
- SEC-HPLC if available

Function

- cell-based proliferation/bioactivity assay if available
-

11. Troubleshooting

Problem: rhG-CSF lost during casein precipitation

Possible reasons:

- rhG-CSF precipitates near pH 4.6
- binds to casein pellet
- unstable at low pH

Fix

- test pH 4.8 instead of 4.6
 - reduce hold time from 20 min to 5–10 min
 - add mild salt during optimization
 - compare acid precipitation with membrane-based casein removal
-

Problem: poor binding to SP column

Possible reasons:

- conductivity too high
- pH too high
- sample overloaded

Fix

- improve diafiltration
 - lower pH slightly to 4.5–4.7
 - reduce load amount
-

Problem: broad or dirty elution peak

Possible reasons:

- too much contaminant protein
- poor wash
- overloaded column

Fix

- strengthen wash
 - reduce load
 - add second ion-exchange step
-

Problem: aggregation

Possible reasons:

- acid exposure
- concentration stress

- freeze-thaw damage

Fix

- shorten low-pH exposure
 - keep everything cold
 - avoid over-concentration
 - add stabilizer after confirming compatibility
-

12. Example one-day workflow

Day 1

- collect milk
- defat
- precipitate casein
- clarify
- UF/DF into acetate buffer
- run cation exchange
- analyze fractions

Day 2

- concentrate pool
 - SEC polishing
 - final analysis
 - formulation and storage
-

13. Simplified version

If you want the **minimum workable SOP**, use this:

1. Centrifuge 1 L milk at 5,000 × g, 30 min, 4°C
 2. Remove fat
 3. Adjust skim milk to pH 4.6
 4. Stir 20 min at 4°C
 5. Centrifuge 10,000 × g, 30 min, 4°C
 6. Collect whey
 7. Filter through 0.45 µm
 8. UF/DF into 20 mM acetate pH 4.8
 9. Load onto SP Sepharose
 10. Wash and elute with NaCl gradient
 11. Pool rhG-CSF fractions
 12. Polish by SEC
 13. Store in acetate or phosphate buffer
-

14. Important note

For **milk-derived rhG-CSF**, one of the first things you should check is whether your product is:

- **glycosylated**
- partially degraded
- present as multiple isoforms

That can change:

- migration on SDS-PAGE
- binding on ion exchange
- elution position
- bioactivity and stability

So this SOP is your **starting process**, not the final optimized one.

Process flow

1 L milk

→ defat by centrifugation

→ precipitate casein at pH 4.6

→ collect whey

→ clarify by filtration

→ UF/DF into acetate buffer

→ SP cation exchange

→ pool target fractions

→ UF concentrate

→ SEC polish

→ formulate/store

I can also prepare a **very detailed batch sheet table** with columns for **step, buffer, volume, pH, temperature, centrifuge speed, expected output, and sampling points**.

sure

Thought for a second

Here is a **detailed batch sheet table** for **purification of recombinant human G-CSF from 1 L milk**.

Batch Sheet

Process: Purification of rhG-CSF from milk
Scale: 1 L starting milk
Process type: Research / lab scale
Product: Recombinant human granulocyte colony-stimulating factor

1. Batch information

Item	Record
Batch ID	
Operator	
Date	
Sample source	
Starting volume	1.0 L
Storage condition before process	
Start time	
End time	

2. Buffer preparation record

Buffer ID	Composition	Target pH	Prepared by	Date	Lot / Notes
Buffer 1	20 mM sodium acetate	4.8			
Buffer 2	20 mM sodium acetate + 100 mM NaCl	4.8			
Buffer 3	20 mM sodium acetate + 1 M NaCl	4.8			
Buffer 4	20 mM sodium phosphate + 150 mM NaCl	7.2			
Buffer 5	10 mM sodium acetate + 150 mM NaCl	4.5			

3. Step-by-step batch sheet

Step No.	Step	Operation details	Buffer / Reagent	Temp.	Time	Volume in
1	Receipt	Receive milk and place on ice / 4°C	None	4°C	—	1000 mL
2	Pre-check	Record appearance, odor, color, clotting	None	4°C	5 min	1000 mL
3	Optional inhibitor	Add protease inhibitor if used	Inhibitor	4°C	5 min	1000 mL
4	Defatting	Centrifuge milk	None	4°C	30 min	1000 mL
5	Fat removal	Remove cream layer carefully	None	4°C	10 min	~850–950 mL
6	Acidification	Adjust skim milk to casein precipitation point	1 N HCl	4°C	10–20 min	~850–950 mL
7	Hold	Hold at pH 4.6	None	4°C	20 min	same
8	Casein removal	Centrifuge acidified milk	None	4°C	30 min	same
9	Supernatant recovery	Recover whey carefully	None	4°C	10 min	~700–850 mL
10	Coarse clarification	Prefilter if needed	Gauze / coarse filter	4°C	5–10 min	~700–850 mL
11	Fine clarification	Filter through 0.45 µm	0.45 µm filter	4°C	10–20 min	~700–850 mL
12	Optional sterile clarification	Filter through 0.22 µm if feasible	0.22 µm filter	4°C	10–20 min	similar

Step No.	Step	Operation details	Buffer / Reagent	Temp.	Time	Volume in
13	UF concentration	Concentrate clarified whey	UF system, 10 kDa MWCO	4–10°C	variable	~700–850 mL
14	Diafiltration	Exchange into binding buffer	Buffer 1	4–10°C	variable	50–100 mL
15	Column equilibration	Equilibrate SP column	Buffer 1	RT / 4–10°C	5–10 CV	—
16	Load	Apply sample to cation-exchange column	Sample in Buffer 1	RT / 4–10°C	variable	~50 mL
17	Wash 1	Wash with equilibration buffer	Buffer 1	RT / 4–10°C	5–10 CV	—
18	Wash 2	Wash with light salt buffer	Buffer 2	RT / 4–10°C	5 CV	—
19	Elution	Elute rhG-CSF by salt gradient or steps	Buffer 3	RT / 4–10°C	10–20 CV	—
20	Fraction screen	Check fractions by SDS-PAGE / ELISA / WB	Assay reagents	—	variable	fractions
21	Pooling	Combine target fractions	None	4°C	10 min	variable
22	UF concentration	Concentrate capture pool	UF system	4–10°C	variable	10–40 mL
23	Buffer exchange	Exchange into SEC buffer	Buffer 4	4–10°C	variable	1–5 mL
24	SEC equilibration	Equilibrate SEC column	Buffer 4	RT / 4–10°C	per column	—
25	SEC run	Inject and separate monomer / aggregates	Buffer 4	RT / 4–10°C	variable	1–5 mL
26	Final fraction analysis	Analyze SEC fractions	Assay reagents	—	variable	fractions
27	Final pooling	Pool final desired fractions	None	4°C	10 min	variable
28	Final filtration	Sterile filter if needed	0.22 µm	4°C	5–10 min	variable
29	Formulation	Exchange to storage buffer if needed	Buffer 5	4°C	variable	variable
30	Aliquot / store	Dispense and freeze/store	Storage tubes	4°C / -80°C	10–20 min	final

4. Sampling plan

Sample ID	Stage	Purpose	Suggested test
S1	Whole milk	Starting reference	ELISA, SDS-PAGE
S2	Skim milk	Check loss during defatting	ELISA
S3	Casein pellet	Check rhG-CSF loss into pellet	Extract + ELISA / WB
S4	Whey supernatant	Recovery after casein removal	ELISA, SDS-PAGE
S5	Clarified whey	Feed quality before UF	SDS-PAGE
S6	UF/DF sample	Column loading readiness	pH, conductivity, ELISA
S7	Column flow-through	Binding check	ELISA, WB
S8	Wash 1	Loss during wash	ELISA
S9	Wash 2	Loss during light-salt wash	ELISA
S10	Elution fractions	Identify target fractions	SDS-PAGE, WB, ELISA
S11	Capture pool	Intermediate purity	SDS-PAGE, concentration
S12	Pre-SEC sample	SEC input check	SDS-PAGE
S13	SEC fractions	Select monomeric target	SDS-PAGE, SEC profile
S14	Final pool	Final product check	SDS-PAGE, WB, ELISA
S15	Formulated product	Release-style research QC	Concentration, purity, activity

5. In-process control sheet

Control point	Target	Actual	Pass / Fail	Comments
Milk kept cold	4°C			
Defatting spin	5,000 × g, 30 min, 4°C			
Casein precipitation pH	4.6			
Casein hold time	20 min			
Casein removal spin	10,000 × g, 30 min, 4°C			
Clarified whey appearance	Low turbidity			
UF final volume	50–100 mL			
UF/DF final pH	4.8			
UF/DF conductivity	Low enough for binding			
CEX load accepted	No major pressure issue			
Target binds column	Minimal target in FT			
SEC monomer peak	Clear major peak			
Final pH	target buffer pH			

Control point	Target	Actual	Pass / Fail	Comments
Final storage condition	4°C or -80°C			

6. Column record

Cation-exchange column

Item	Record
Resin	SP Sepharose or equivalent
Column ID	
Bed volume	
Equilibration buffer	Buffer 1
Load volume	
Load flow rate	
Wash 1 volume	
Wash 2 volume	
Elution type	Gradient / Step
Fraction size	
Pressure notes	

Size-exclusion column

Item	Record
Resin / Column type	
Column ID	
Equilibration buffer	Buffer 4
Injection volume	
Flow rate	
Fraction size	
Main peak fraction numbers	
Pressure notes	

7. Analytical results sheet

Sample ID	Test	Result	Unit	Comments
S1	rhG-CSF ELISA			
S4	rhG-CSF ELISA			

Sample ID	Test	Result	Unit	Comments
S7	rhG-CSF ELISA			
S10	SDS-PAGE			
S10	Western blot			
S11	Total protein			
S14	Total protein			
S14	Purity estimate		%	
S14	rhG-CSF ELISA			
S15	Activity assay			

8. Yield and recovery sheet

Stage	Volume (mL)	rhG-CSF conc.	Total rhG-CSF	Step recovery %	Cumulative recovery %	Notes
Whole milk	1000			100	100	
Skim milk						
Whey supernatant						
Clarified whey						
UF/DF retentate						
CEX pool						
SEC pool						
Final formulated product						

9. Deviation / troubleshooting log

Step	Issue observed	Possible cause	Action taken	Outcome
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10. Common acceptance guide

Step	Acceptable sign
Defatting	Clear cream layer separated
Casein precipitation	Visible curd/pellet formation near pH 4.6
Clarification	Whey becomes less turbid
UF/DF	Stable concentration, manageable membrane pressure

Step	Acceptable sign
CEX	Target mostly absent from flow-through
Elution	Distinct protein-containing fractions
SEC	Clean main monomer peak
Final pool	Strong target band, reduced impurity bands

11. Very practical notes

Casein step

This is the most critical early checkpoint.
If rhG-CSF is lost here, test:

- pH 4.6 vs 4.8
- hold 20 min vs 5–10 min
- pellet extraction for target recovery

CEX step

If target appears in flow-through, likely:

- conductivity too high
- pH too high
- column overloaded

SEC step

Use it mainly as a polishing step, not a capture step.

12. Compact printable version

Batch ID:
Date:
Operator:
Start material: 1 L milk

1. Defat: 5,000 × g, 30 min, 4°C

2. Remove fat layer

3. Adjust skim milk to pH 4.6

4. Hold 20 min at 4°C

5. Centrifuge 10,000 × g, 30 min, 4°C

6. Collect whey

7. Filter 0.45 μm

8. UF/DF into 20 mM acetate pH 4.8

9. Load SP column

10. Wash with Buffer 1 and Buffer 2

11. Elute with Buffer 3

12. Pool target fractions

13. UF concentrate

14. Exchange into Buffer 4

15. Run SEC

16. Pool monomer fractions

I can turn this into a **Word document**, **Excel sheet**, or **PDF template** with clean formatting.

